Circular motion

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Relate inward net force to a change in velocity direction.

SUCCESS CRITERIA - I can...

1. Draw tangent velocity and inward acceleration.
2. Use v^2/r or a supplied acceleration.
3. Identify the real interaction providing centripetal force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A projectile's velocity is (9, -12) m/s. Find its speed. What information would locate it?

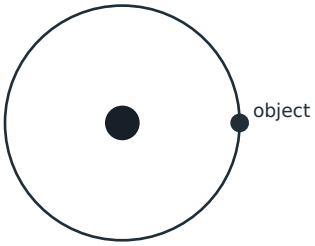
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $a_c = v^2/r$; $F_{net} = mv^2/r$.
c means centripetal: toward the center. This is the net inward force, not an extra force.

Worked reasoning: A 2 kg object moves at 3 m/s in a circle of radius 3 m. Inward acceleration is $v^2/r = 3\ m/s^2$; net force is 6 N inward.

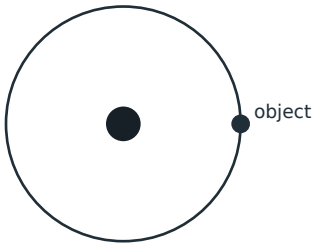
Your turn: Draw tangent velocity and inward net force at two different points.



Marzano check - circle: ① ② ③

03 Read and interpret the model

Add velocity and acceleration arrows to the marked point on the circle. Name one interaction that could provide the inward force.

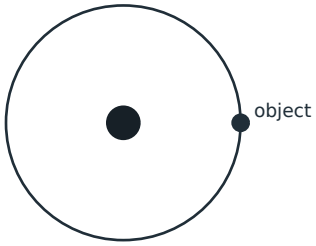


The center is the attracting body; the small dot on the circle is the orbiting object.

Marzano check - circle: ① ② ③

04 Apply the physics

A 0.50 kg puck on a string moves at 4 m/s in a horizontal circle of radius 2 m. Draw velocity and net-force arrows. Calculate inward acceleration and net force. Then predict both quantities at twice the speed and the same radius.



Marzano check - circle: ① ② ③

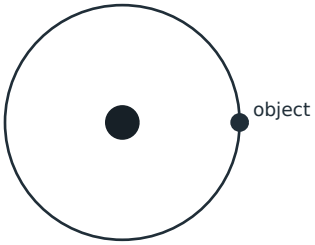
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Continue down each column.

05 Explain your reasoning

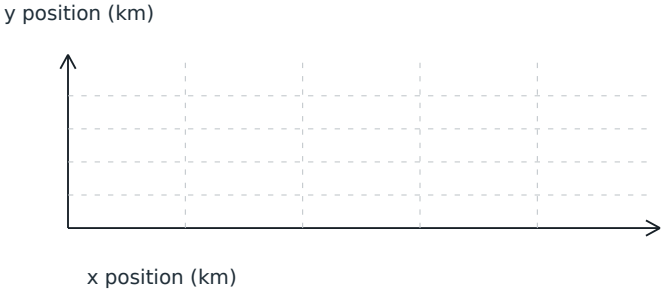
A string breaks while a ball moves in a circle. Neglect other forces over the next instant. Sketch its initial path after release.



Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

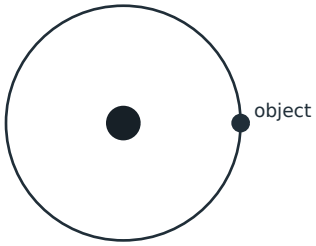
A simplified model predicts position (-80,100) km with bounds ±8 km in x and ±10 km in y. Write a short briefing giving both intervals and two reasons it is not a real asteroid collision forecast.
Label the given and predicted quantities on the model.



Marzano check - circle: 1 2 3

07 Exit check

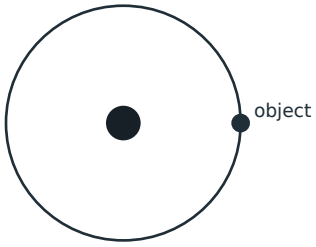
At fixed speed and mass, double circle radius. How do acceleration and net force change? Explain from equations. Label the radius and show velocity and inward acceleration on the circle.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

The maximum horizontal force on a 0.5 kg object is 8 N. For a 1 m circle, find the maximum speed. Then double the radius and find the new limit. Label the radius and show velocity and inward acceleration on the circle.



Marzano check - circle: 1 2 3